LAB-10

Title- Performance Analysis of Ensemble Learning Techniques Using Random Forest or Boosting

Objectives

1. To understand the concept of **ensemble learning**.
2. To understand the working principle of **Random Forest and Boosting techniques**.
3. To implement an ensemble learning technique such as **Random Forest or Boosting** on a suitable open-access dataset.
4. To implement a **baseline machine learning model** for comparison.
5. To compare the performance of ensemble and baseline models using appropriate evaluation metrics.
6. To analyze the advantages of ensemble learning in improving prediction performance.

Problem Statement- Analyze the performance of ensemble learning techniques such as Random Forest or Boosting on a suitable open-access dataset in comparison with a baseline machine learning model.

Theory

Ensemble Learning

Ensemble learning is a machine learning technique that combines the predictions of multiple machine learning models to obtain a better and more reliable prediction. The basic idea is: Multiple models working together can often perform better than a single model. Ensemble learning can improve accuracy, stability, and generalization of machine learning models.

Common ensemble techniques include:

- Random Forest
- AdaBoost
- Gradient Boosting
- XGBoost
- Voting
- Bagging

Random Forest

Random Forest is an ensemble learning algorithm that combines multiple Decision Trees. Each decision tree is trained using different subsets of the training data and features. The predictions from all the trees are then combined. For classification, the final prediction is generally based on majority voting.

Boosting

Boosting is an ensemble technique in which multiple weak learners are trained sequentially.

Each new model tries to improve the mistakes made by the previous models. Common boosting algorithms include:

- AdaBoost
- Gradient Boosting
- XGBoost

The final model combines the predictions of all the weak learners to produce a stronger model.

Baseline Model

A **baseline model** is a simple machine learning model used as a reference for comparison. For example, for a classification problem, we can use:

- Logistic Regression
- Decision Tree
- K-Nearest Neighbors (KNN)

Conclusion

The ensemble learning technique, such as Random Forest or Boosting, is successfully implemented on a suitable open-access dataset and compared with a baseline machine learning model. The performance of the models is evaluated using appropriate metrics. The comparison helps to understand whether combining multiple learners provides better prediction accuracy, stability, and generalization than a single baseline model.

Viva Questions

1. What is ensemble learning?
2. Why is ensemble learning used?
3. What is Random Forest?
4. How does Random Forest work?
5. What is a Decision Tree?
6. What is the difference between a Decision Tree and Random Forest?
7. What is Boosting?
8. How does Boosting work?
9. What is a weak learner?
10. What is the difference between Bagging and Boosting?
11. What is majority voting in Random Forest?
12. How does Random Forest reduce overfitting?
13. What is feature importance in Random Forest?
14. What is the difference between Random Forest and Boosting?
15. What are the advantages and limitations of ensemble learning?